Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/28/2026
PI Name:	Jones, Christopher W. Ph.D.		
Project Title:	Mars Adaptive Training Integrative Knowledge System (MATRIKS) to Improve Operational Performance and Its Neural Basis for Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHP, OMNIBUS3 Human Research Program: Human Health & Performance Appendix E; Omnibus3-Appendix F
Start Date:	12/17/2021	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per A. Beitman/HFBP. Original end date was 09/30/2027 (Ed., 8/14/25).		
Key Personnel Changes/Previous PI:	N/A		

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Task Description:	With prolonged mission durations, spaceflight crews will become increasingly dependent on onboard technologies for knowledge acquisition and maintenance. It is expected that not all skills and knowledge required for these missions can be retained and retrieved based on pre-mission training alone. Limited and delayed communication will constrain support from Mission Control and less assistance will be available when performing mission-critical operations, such as post-landing operations. With the present project we will target NASA's particular interest in developing and assessing an adaptive, just-in-time countermeasure that will consolidate and improve skills that are most relevant to space flight operations. To achieve this aim, NASA established a Virtual NASA Specialized Center of Research (VNSCOR) referred to as the "Mars Adaptive Training Integrative Knowledge System (MATRIKS)", which is comprised of the following projects: (1) "Trinity – Multi-Environment Virtual Training for Long Duration Exploration Missions", (PI: A. Anderson, CU Boulder); (2) "Morpheus – A Haptic Sensory Supplement to Optimize In-Flight Adaptive Training for Human Control of Spacecraft Robotic Arms" (PI: S. Robinson, UC Davis); and the present project "Neo – Adaptive Training integrative knowledge System to Improve Operational Performance and its Neural Basis for Spaceflight" (PI: C.W. Jones, UPenn). Neo leverages a validated workstation, Six Degrees of Freedom (6DF), that simulates a rendezvous and docking maneuver using real spacecraft flight dynamics. It is designed to (1) train and improve sensorimotor skills relevant for inflight and post-landing operational tasks; (2) feature an autonomous and adaptive training approach that does not rely on feedback from flight operations on the ground; (3) maximize the transfer of mission-relevant motor skills; (4) allow the assessment of the neural circuitry underlying the task; and (5) deliver the training in a motivating and meaningful way to astronauts. Neo comprises two overarching aims: First, we will identify the neural circuitry underlying spaceflight relevant tasks by performing a subset of 6DF tasks during functional magnetic resonance imaging (fMRI) in up to N=30 healthy adults with varying levels of 6DF experience. Second, we will integrate the proposed 6DF autonomous intelligent tutor system in an additive manner with a haptic feedback intervention (Morpheus) and multi-environment virtual trainer (Trinity) as part of the VNSCOR MATRIKS. It is expected that Neo, Morpheus, and Trinity mutually complement each other to produce an effective countermeasure tool to acquire and retain operational skills that are critical for exploration-class missions. To assess the efficacy of this combined effort, the VNSCOR MATRIKS will collect data in one 45-day HERA campaign of N=16 crewmembers (four missions with N=4 crewmembers each). The primary goal is to identify changes in operational performance, indexed by the Robotic On-board Trainer for Research (ROBoT-r), in response to MATRIKS. As part of Neo, we will also examine whether the effects of MATRIKS transfer to general cognitive functioning (indexed by performance on the Cognition test battery), distinctive visuospatial abilities critical for performing telerobotic tasks (indexed by performance on the Spatial Cognition battery), and brain structure and neural circuitry of key brain networks relevant for spaceflight-related performance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected significance of this project relates to its relevance for facilitating effective countermeasure tools to acquire and retain operational skills that are critical for exploration class missions. At the conclusion of the research, we will have defined and demonstrated the use of a neuroscience-based, adaptive training integrative knowledge system to potentially mitigate visuospatial and sensorimotor brain changes associated with prolonged isolation and confinement to reduce the likelihood or impact of potential decrements in human performance capabilities during long-duration space missions. Together, these data will help mission planners to ensure safe and successful space exploration class missions. It is possible that the results from this project also translate to situations on Earth where fine motor skills are essential, such as robot-assisted surgery.
Task Progress:	Proposed work for specific aim 1 was completed. A pilot study of N=30 healthy adults was conducted at the German Aerospace Center (DLR) to optimize the functional neuroimaging protocol for the Six-Degrees of Freedom (6DF) task and characterize the neural basis of complex operational performance. Data collection in the DLR study was successful, with 100% of the expected data collected. Data were assessed for completeness and underwent quality control prior to processing. The following data types and quantities were collected: 1. MRI: Measuring neural anatomy and task-dependent functional activation, a total of N=30 structural and N=30 functional MRI scans were collected. 2. 6DF: Measuring operationally-relevant performance, a total of N=30 6DF sessions were collected, consisting of N=960 individual 6DF trials. 3. Cognition: Measuring cognitive performance, a total of N=60 Cognition test batteries were collected, consisting of N=600 Cognition tests. 4. Spatial Cognition: Measuring spatial cognition performance using eight separate tests indexing navigation search and strategy, as well as spatial orienting and updating, a total of N=240 spatial cognition tests were collected. 5. Surveys: Four different surveys, totaling N=120 surveys, were collected on participant demographics, handedness (Edinburgh Handedness Inventory), computer experience, and sense of direction. Analyses of operational and cognitive performance, as well as the neural activation during operational performance, were completed. We demonstrated that performing the 6DF task resulted in specific activation/deactivation patterns consistent with the broader literature. We also found that neural suppression on more difficult operational maneuvers was related to cognitive performance, including speed on Cognition and accuracy on Spatial Cognition.

Bibliography Type:	Description: (Last Updated: 02/18/2026)
Abstracts for Journals and Proceedings	Jones CW, Basner M, Bell S, Brauns K, Dinges DF, Fischer F, Friedl-Werner A, Gerlach DA, Gur RC, Ivkovic V, Johannes B, Kuhn S, Muhl C, Piechowski S, Roalf D, Strangman G, Stahn AC. "NEO – MARS Adaptive Training Integrative Knowledge System (MATRIKS) to improve operational performance and its neural basis for spaceflight." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025